TWO MORPHISMS AGREEING ON A SCHEME THEORETICALLY DENSE OPEN SUBSPACE OF A SEPARATED TARGET COINCIDE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short with no references at all; the two ingredients are the closedness of the equalizer and scheme theoretic density of U.

Editorial note. The counted spans carry no external references, so no navigation stubs are needed beyond the ones implied by the statement wording.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Separatedness of Y over B means the diagonal is a closed immersion. used at proof lines 3597-3599. Makes the equalizer Z a closed subspace of X.
- Scheme theoretic density tested by the closure of an open subspace. used at proof lines 3599-3601. Forces Z = X from U contained in Z.
- The equalizer of two morphisms as a fibre product along the diagonal. used at proof lines 3598-3599. Identifies Z and its closedness.

Lemma 0.2. *Let S be a scheme. Let B be an algebraic space over S. Let $f, g : X \to Y$ be morphisms of algebraic spaces over B. Let $U \subset X$ be an open subspace such that $f|_U = g|_U$. If the scheme theoretic closure of U in X is X and $Y \to B$ is separated, then $f = g$.*

Proof. As $Y \to B$ is separated the fibre product $Y \times_{\Delta, Y \times_B Y, (f,g)} X$ is a closed subspace $Z \subset X$. As $f|_U = g|_U$ we see that $U \subset Z$. Hence $Z = X$ as U is assumed scheme theoretically dense in X. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.